Kudeki

Exam 2

Thu, July 12, 2018 — 11 AM - 11:50

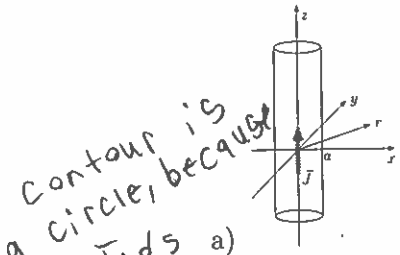
Name:	Solution
Section:	

Please clearly PRINT your name in CAPITAL LETTERS and circle your section in the above boxes.

This is a closed book exam and calculators are not allowed. You are allowed two formula sheets of 8.5 by 11 inch dimensions — both sides of the sheets may be used. Please show all your work and make sure to include your reasoning for each answer. All answers should include units wherever appropriate.

Problem 1 (25 points)	
Problem 2 (25 points)	
Problem 3 (25 points)	
TOTAL (75 points)	

1. **DC magnetic field problems:**



The current density $\mathbf{J}(z,y,z)$ in the cylinder shown in the left is given as $\hat{z}/\pi$ A/m^2 for $r \equiv \sqrt{x^2+y^2} < a$ and zero otherwise — this is essentially a cylindrical shaped volumetric current centered about the z-axis and the cylinder has a radius a. This volumetric current density produces a DC magnetic field $\mathbf{B} = B_\phi(r)\hat{\phi}$ surrounding the z-axis in the azimuthal direction $\hat{\phi}$, just like an infinite thin wire with some current I flowing along the z-axis.

Contour is a circle, because $\int J \cdot ds$ a) only $\neq 0$ through the cap



i. (6 pts) Using the integral form of Ampere's law show that $B_\phi(r) = \frac{\mu_o r}{2\pi}$ if $r < a$. Your answer should have a picture illustrating what shape of an integration contour you are using in applying Ampere's law to the solution of this problem and explain your reasoning.

$\oint \frac{B}{\mu_0} \cdot dl = \int J \cdot ds$

$\frac{B(r)}{\mu_0} 2\pi r = |J| \pi r^2$, $\quad B(r) = \frac{|J| \mu_0 \pi r^2}{2\pi r} = \frac{\mu_0 r}{2\pi}$ ✓

ii. (5 pts) Using the integral form of Ampere's law determine $B_\phi(r)$ for $r > a$.

$\oint \frac{B}{\mu_0} \cdot dl = \int J \cdot ds$

$\frac{B(r)}{\mu_0} 2\pi r = |J| \pi a^2$,

$B(r) = \frac{\mu_0 a^2}{2\pi r}$ ✓

iii. (2 pts) What is H_ϕ if $a = \pi$ m and $r = a$?

$H = \frac{r}{2\pi} = \frac{1}{2} \frac{A}{m}$ ✓

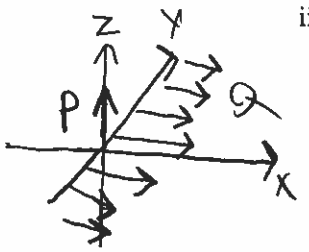
iv. (2 pts) What is H_ϕ if $a = 1$ m and $r = 2a$?

$H = \frac{a^2}{2\pi r} = \frac{1}{2\pi(2)} = \frac{1}{4\pi} \frac{A}{m}$ ✓

b) Consider the DC current density function $\mathbf{J}(x,y,z) = P\delta(x)\delta(y)\,\hat{z} + Q\delta(z)\,\hat{x}$ A/m^2 where coordinates x, y, and z are measured in meter units.

i. (2 pts) What are the units of parameters P and Q in the MKSA system? Justify your answer.

$\delta(x)$ [=] $\frac{1}{m}$, so P [=] A, Q [=] $\frac{A}{m}$ ✓ ✓



ii. (3 pts) Sketch a diagram depicting the current system described by $\mathbf{J}(x,y,z)$ in the Cartesian coordinate system indicating the locations and directions of current flows. ✓

iii. (5 pts) If $P = Q = 1$ in MKSA units determine $\mathbf{H}(1,0,1)$ caused by $\mathbf{J}(x,y,z)$? Show your work.

$H = \frac{P}{2\pi x} - \frac{Q}{2} = \frac{1}{2\pi} - \frac{1}{2} = \frac{2-\pi}{2\pi} \hat{y} \frac{A}{m}$ ✓

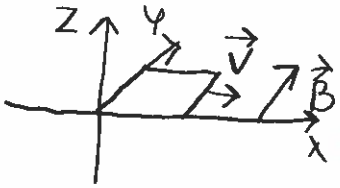
by RHR, H from P is in $\hat{y}$, Q is $-\hat{y}$

2. **Faraday effect and emf problem:** A conducting square wire loop of dimensions 1 m × 1 m is moved with velocity $\mathbf{v} = 4\hat{x}$ m/s in a region where the background magnetostatic field is described by

$$\mathbf{B}(x,y,z) = \hat{y}\, 25 \times 10^{-6}(1 + x/L)\ \text{T},$$

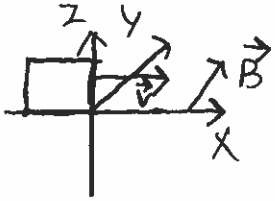
where $L = 1000$ m. The center of the loop coincides with the origin $(x,y,z) = (0,0,0)$ at $t = 0$.

a) (4 pts) Show a possible orientation for the loop if no current is induced in the loop in motion? Show with a diagram.



The loop is Parallel to $\vec{B}$, so $\int B \cdot ds = 0$

b) (4 pts) What orientation for the loop maximizes the current flow through the loop? Show with a diagram.



loop is Perpendicular to B

c) (8 pts) Obtain an expression for the flux $\Psi(t)$ of the loop in motion for $t > 0$ for the orientation determined in part (b). Show the circulation direction C consistent with your $\Psi(t)$ in the diagram in part (b). Since the loop dimension of 1 m is much smaller than L the magnetic field across the loop can be approximated by the field value at the loop center.

$$\Psi(t) = \int \hat{y}\left(25\times10^{-6}\left(1+\frac{x(t)}{L}\right)\right)\cdot ds \quad \text{wb}$$

$$\Psi(t) = 25\times10^{-6}\left(1+\frac{4t}{L}\right)\ \text{V}\cdot\text{s}$$

d) (6 pts) Obtain an expression for the induced emf $\mathcal{E}(t)$ of the loop in motion for $t > 0$ from $\Psi(t)$ of part (c).

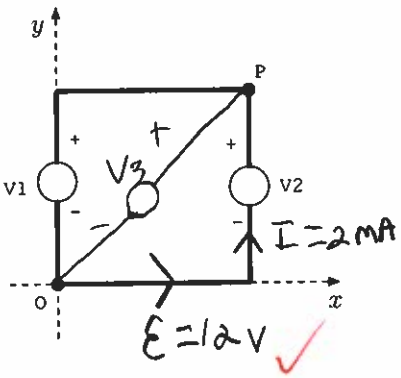
$$\mathcal{E} = -\frac{d}{dt}\Psi(t) = -25\times10^{-6}\,\frac{4}{L} = -10^{2}\cdot10^{-6}\cdot10^{-3}$$

$$= -10^{-7}\ \text{V}$$

e) (3 pts) What is the magnitude of the loop current if the loop resistance is 2 Ω?

$$|I| = \frac{|\mathcal{E}|}{R} = \frac{10^{-7}}{2}\ \text{A}$$

3.



Voltmeters V1 and V2 are connected between points O=(0,0) and P=(2,2) m on the xy-plane (as shown in the figure on the left) along the sides of a 2x2 m square. A uniform magnetic field $\mathbf{B} = -\hat{z}3t$ Wb/m^2 is linking the square.

a) (3 pts) Using $d\mathbf{S} = \hat{z}dxdy$ determine the magnetic flux $\Psi(t) = \int_S \mathbf{B} \cdot d\mathbf{S}$ linking the square loop formed by the connected voltmeters.

$\Psi(t) = \int_S (-\hat{z}\,3t) \cdot (\hat{z}\,dx\,dy) = -3t\,(2)^2$

$= -12t$ V·s ✓

b) (4 pts) Determine the strength and direction of the emf $\mathcal{E}$ for the square loop using $\Psi(t)$ from part (a) — indicate your computed emf on the diagram explicitly with an arrow and value.

$\mathcal{E} = -\frac{d}{dt}\Psi(t) = 12$ V ✓

c) (3 pts) Suppose the internal resistance of voltmeters V1 and V2 are $R_1 = R_2 = 3\,\text{k}\Omega$. In that case what would be the loop current I? Indicate I on the diagram explicitly with an arrow and a value. **Hint:** in this calculation the voltmeters are treated as resistors R_1 and R_2, respectively.

$I = \frac{\mathcal{E}}{R} = \frac{12}{6\text{k}} = 2\,\text{mA}$ ✓

d) (10 pts) What readings would V1 and V2 display (paying attention to polarities of the voltmeter leads)? Justify your answer. **Hint:** each voltmeter displays the voltage drop $\int \mathbf{E} \cdot d\mathbf{l}$ from its + probe tip to its - probe tip!

$V_2 = -I\,R_2 = -(2\text{mA})(3\text{k}\Omega) = -6$ V ✓

$V_1 = I\,R_1 = (2\text{mA})(3\text{k}\Omega) = 6$ V ✓

e) (5 pts) If a third voltmeter V3 with an internal resistance of $R_3 = 3000\,\text{k}\Omega$ were connected between O and P diagonally across the 2x2 m square, what would it be its voltage reading assuming its + probe is connected to P? Explain! **Hint:** since the internal resistance of V3 is very high (compared to R_1 and R_2) you may assume that loop current I calculated in part (b) will not be changed after connecting V3 as indicated.

$\frac{\mathcal{E}_0}{2} + V_2 - V_3 = 0$

$\frac{12}{2} - 6 = V_3,\ V_3 = 0$ V ✓

Placing the probe across the diagonal gives a reading where $\mathcal{E} = \frac{\mathcal{E}_0}{2}$, where $\mathcal{E}_0 = 12$